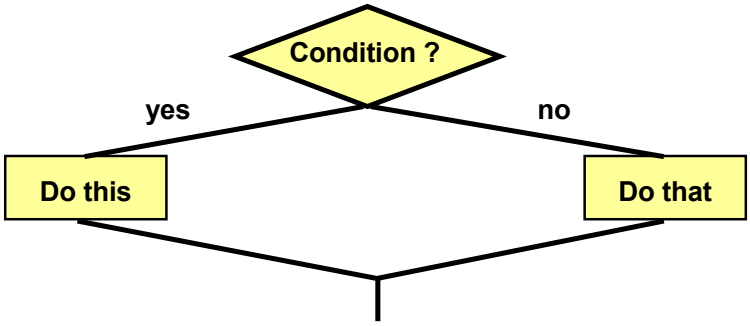
3 Selections and Loops

3.1 Introduction

In this chapter you will learn to incorporate intelligence into your programs, i.e. the program can do different things depending on different conditions (selections). You will also learn how to repeat certain tasks a specific number of times or until a specific condition is fulfilled (iteration, loop). We will introduce new symbols in our JSP graphs to illustrate selections and loops.

3.2 Selection

A selection situation can be illustrated by the following figure:



If the condition is fulfilled (yes option) the program will do one thing, else (no option) another thing.

3.3 if statement

The selection situation is in C++ coded according to the following syntax:

```
if (condition)
  statement1;
else
  statement2;
```

The keyword `if` introduces the if statement. The condition is put within parentheses. If the condition is true statement1 will be performed, otherwise statement2. Here is a code example:

```
if (a>b)
  greatest = a;
else
  greatest = b;
```

The values of two variables are compared. If a is greater than b, the variable greatest will get a's value. Otherwise, i.e. if b is greater than or equal to a, greatest will get b's value. The result from this code section is that the variable greatest will contain the greatest of a and b.

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Sometimes you might want to perform more than one statement for an option. Then you must surround the statements with curly brackets:

```
if (condition)
{
   statements
   …
}
else
{
   statements
   …
}
```

If the condition is true all statements in the first code section will be executed, otherwise all statements in the second code section will be executed. Example:

```
if (a>b)
{
   greatest = a;
   cout << "a is greatest";
}
else
{
   greatest = b;
   cout << "b is greatest";
}
```

If a is greater than b, the variable greatest will get a's value and the text "a is greatest" will be printed. Otherwise the variable greatest will get b's value and the text "b is greatest" will be printed.

Sometimes you don't want to do anything at all in the else case. Then the else section is simply omitted like in the following example:

```
if (sum>1000)
{
   dDiscPercent = 20;
   cout << "You will get 20 % discount";
}
```

If the variable sum is greater than 1000 the variable dDiscPercent will get the value 20 and the text "You will get 20% discount" will be printed. Otherwise nothing will be executed and the program goes on with the statements after the last curly bracket.

3.4 Price Calculation Program

We will now create a program that calculates the total price of a product. The user is supposed to enter quantity and price per unit of the product. If the total exceeds 500:- you will get 10 % discount, otherwise 0 %. We start with a JSP graph:

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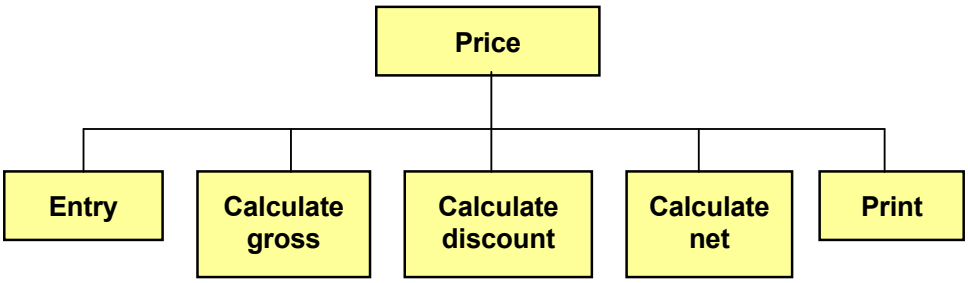
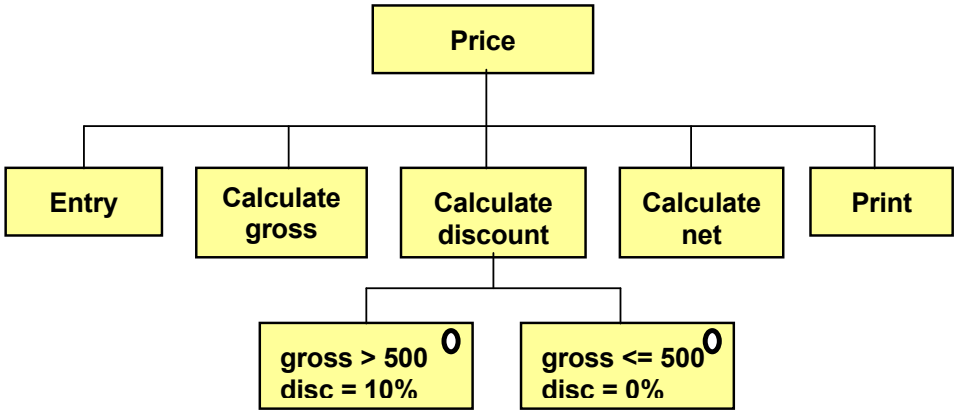


All boxes except "Calculate discount" are rather simple to code. "Calculate discount" requires a closer examination. It has a condition included which says that the discount is different depending on whether gross is less or greater than 500. We'll break down that box:



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A conditional situation in JSP is identified by a ring in the upper right corner of the box. That implies that only one of the boxes will be executed. Here is the code:

```
#include <iostream.h>
void main()
{
  const double dLimit = 500;
  int iNo;
  double dUnitPrice, dGross, dNet, dDisc;
  cout << "Specify quantity and unit price";
  cin >> iNo >> dUnitPrice;
  dGross = iNo * dUnitPrice;
  if (dGross > dLimit)
    dDisc = 10;
    else
    dDisc = 0;
  dNet = (100- dDisc) * dGross / 100;
  cout << "Total price: " << dNet;
}
```

The declaration shows a constant dLimit, which later is used to check the gross value. The variable iNo is used to store the entered quantity and dUnitPrice is used for the entered unit price.

It is common among programmers to use one or a few characters in the beginning of the variable name to signify the data type of the variable. The variable iNo has first character I (integer), and the variable dUnitPrice has d (double).

After data entry the gross is calculated by multiplying the two entered values (quantity * unit price). That value is stored in the variable dGross.

The if statement then checks the value of dGross. If greater than dLimit (i.e. 500) the variable dDisc will get the value 10, otherwise 0. dDisc contains the discount percent to be applied.

The net is then calculated by subtracting the discount percent from 100, which then is multiplied by dGross and divided by 100 (to compensate for the percent value).

Finally the total price is printed.

3.5 Comparison Operators

In the if statements in previous example codes we have so far only used the comparison operator > (greater than). Here is a list of all comparison operators:

```
<   less than
>   greater than
<=  less than or equal to
```


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```
>=  greater than or equal to
==  equal to
!=  not equal to
```

3.6 Even or Odd

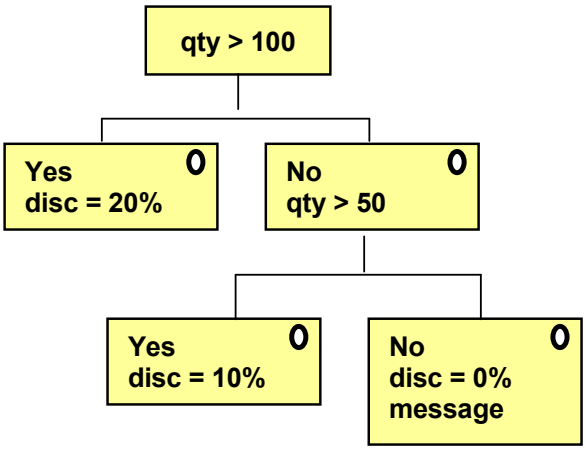
In some situations you will need to check whether a number is evenly dividable by another number. Then the modulus operator % is used. Below are some code examples of how to check whether a number is odd or even, i.e. evenly dividable by 2.

```
//If iNo is even, the remainder of the integer
//division by 2 equals 0:
if (iNo%2 == 0)
   cout >> "The number is even";

//If the remainder of the integer division by 2
//does not equal 0, the number is not dividable
//by 2:
if (iNo%2 != 0)
   cout >> "The number is odd";

//Short way of codeing. An expression not equal
//to 0 is regarded as false, otherwise true.
//If iNo is odd, iNo%2 gives a non zero value:
if (iNo%2)
   cout >> "The number is odd";
```

3.7 else if

We will now study an example of a more complicated situation. Suppose the following conditions prevail:

If a customer buys more than 100 pieces, he will get 20% discount. Otherwise if the quantity exceeds 50, i.e. lies in the interval 50-100, he will get 10%. Otherwise, i.e. if the quantity is below 50, no discount is given. The situation is shown by the following JSP graph:



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The code for this will be:

```
if (iNo>100)
   dDisc = 20;
else if (iNo>50)
   dDisc = 10;
else
{
   dDisc = 0;
   cout << "No discount";
}
```

Here we use the keyword else if.

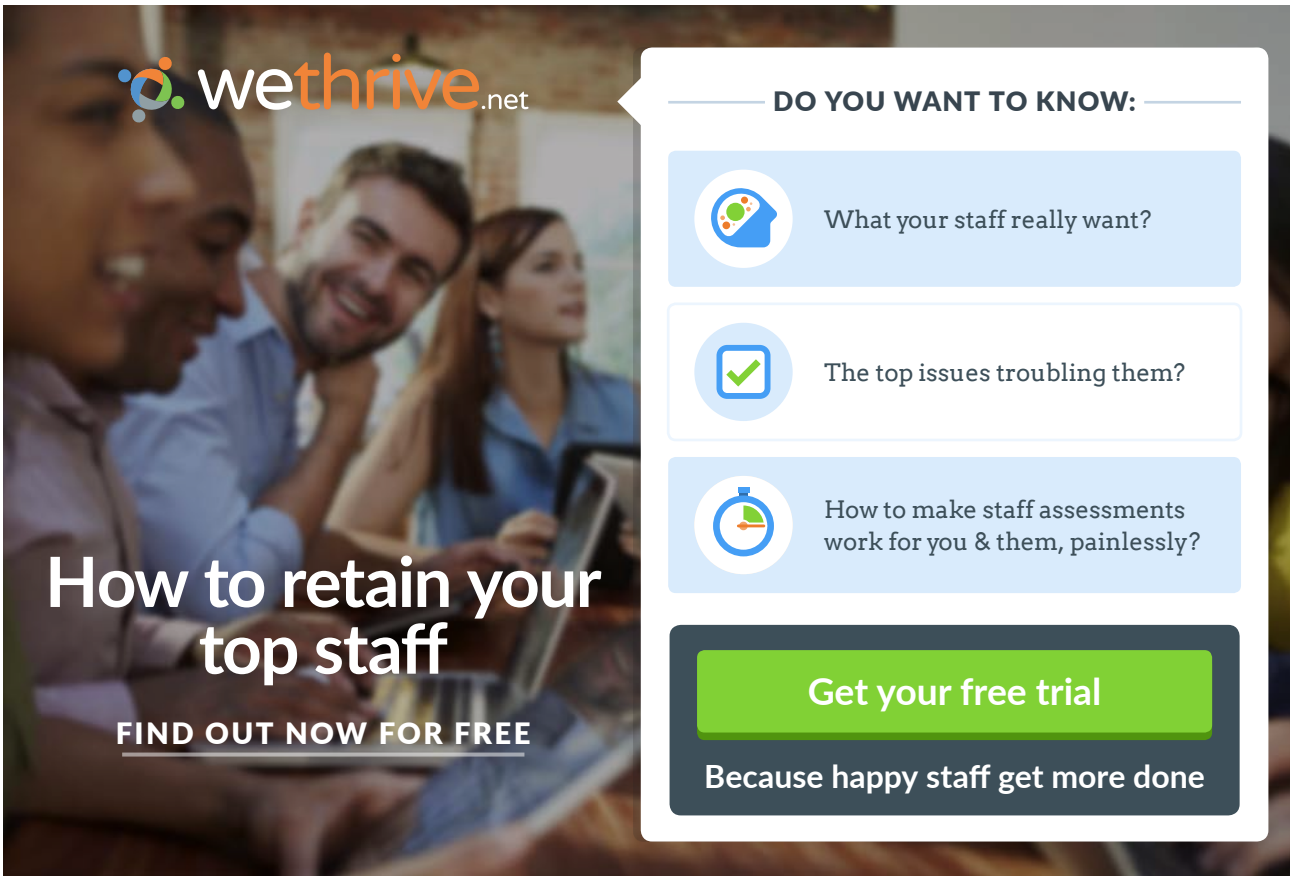
You can use any number of else if-s to cover many conditional cases.

3.8 and (&&), or (||)

The situation with different discount percentages for different quantity intervals can be solved in another way, namely by combining two conditions. In common English it can be expressed like this:

If the quantity is less than 100 and the quantity is greater than 50, the customer will get 10% discount.

Here we combine two conditions:
- If the quantity is less than 100



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and
- and the quantity is greater than 50

The combination of the conditions means that the quantity lies in the interval 50-100. Both conditions must be fulfilled in order to get 10%. The conditions are combined with "and" which is a logical operator. It is written && in C++. The code will then be:

```
if (iNo<100 && iNo>50)
   dDisc = 10;
```

Suppose the situation is this:

If the quantity is greater than 100 or the total order sum is greater than 1000, the customer will get 20% discount.

Here we combine the conditions:
- If the quantity is greater than 100
eller
- or the total order sum is greater than 1000

In both cases the customer has bought so much that he will get 20% discount. One of the conditions is sufficient to get that discount. The conditions are combined with the logic operator "or", which is written || in C++. The code for this situation will be:

```
if (iNo>100 || dSum>1000)
   dDisc = 20;
```

3.9 Conditional Input

In many situations you cannot predict what a user is going to enter. It might happen that the user enters characters when the program expects integers, or that he does not enter anything at all but just press Enter. Then you can use conditional input:

```
if (cin >> iNo)
   ...
```

To understand how this code works you must know that cin is a function that returns a value. If reading of the value to the variable iNo succeeded, the return value from cin is true, otherwise false. Here is a code section that shows how it can be used:

```
cout << "Specify quantity: ";
if (cin >> iNo)
   dTotal = iNo * dUnitPrice;
else
{
   cout << "Input error";
   cin.clear();
   cin.get();
}
```

First we prompt the user for a quantity. The the program halts (cin) and waits for a vlue. If data entry turned out well, the whole condition is true, and the total price is calculated.

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If the data entry failed, i.e. if the user entered letters or just pressed Enter, the condition is false and the statements after else are executed. The user will get a message about input error, the keayboard buffer is cleared (cin.clear()) and the next character in the queue is read (cin.get()). This clean-up procedure must be performed to be able to enter new values to the program.

3.10 The switch statement

In addition to the if statement there is another tool that allows you to perform different tasks depending on the circumstances. The tool is called switch statement and is best accommodated to the situation when checking a value against several alternatives. A good example is a menu where the user enters a menu option (1, 2, 3 ... or A, B, C ...) to make the program do different things depending on the user's choice.

The switch statement has the following syntax:

```
switch (opt)
{
  case 'A':
    //statements
    break;
  case 'B':
    //statements
    break;
  …
  default:
    cout << "Wrong choice";
    break;
}
```

First comes the keyword switch. Within parenthesis after switch there is the variable to be checked. It is checked against the values after the different case keywords. If for instance opt has the value 'A', i.e. opt is a char variable in the example above, then the statements below case 'A' are executed. Note that the keyword break must be found at the end of each case block. If break is omitted, the program will continue into the next case block. Note also that there must be a colon (:) after each case line. The default block takes care of all other options, i.e. if the variable opt does not contain any of the values 'A', 'B' etc. then the statements in the default block will be executed. The entire switch block should be surrounded by curly brackets.

3.11 Menu Program

We will now write a menu program that illustrates how the switch statement can be used. First, the user is prompted for two numbers, and then a menu is displayed where the user can select whether to view the greatest, the least, or the average of the two numbers. The screen will look like this:

```
Enter 2 numbers:  7 5
1.  Greatest
2.  Least
3.  Average
Select:
```


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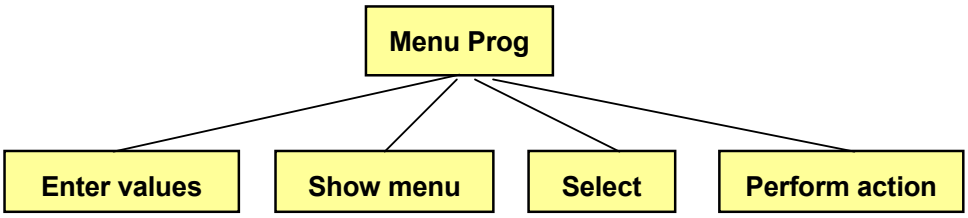
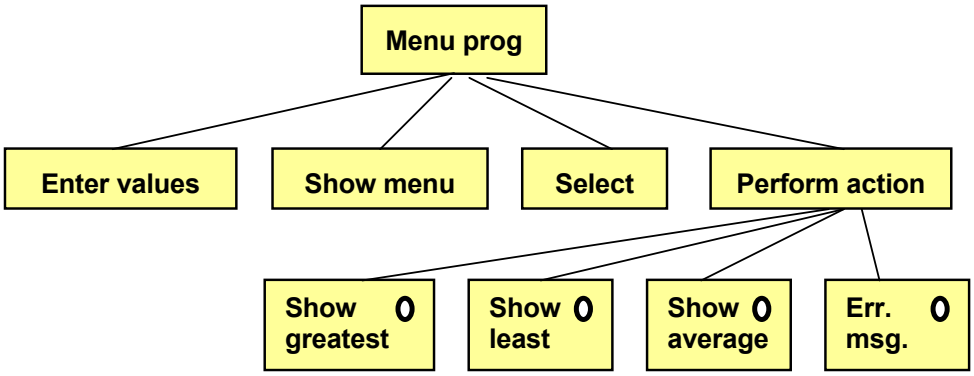
First, the user has entered the numbers 7 and 5. Then a menu is displayed where the user has to select 1, 2 or 3, depending on what he wants to view.

We will first draw a JSP graph that explains the process:



First, the user enters two numbers. Then the menu is displayed on the screen and the user selects an option. Finally the requested action is performed.

The requested action can be one of four options, so we break down the box ”Perform action”:



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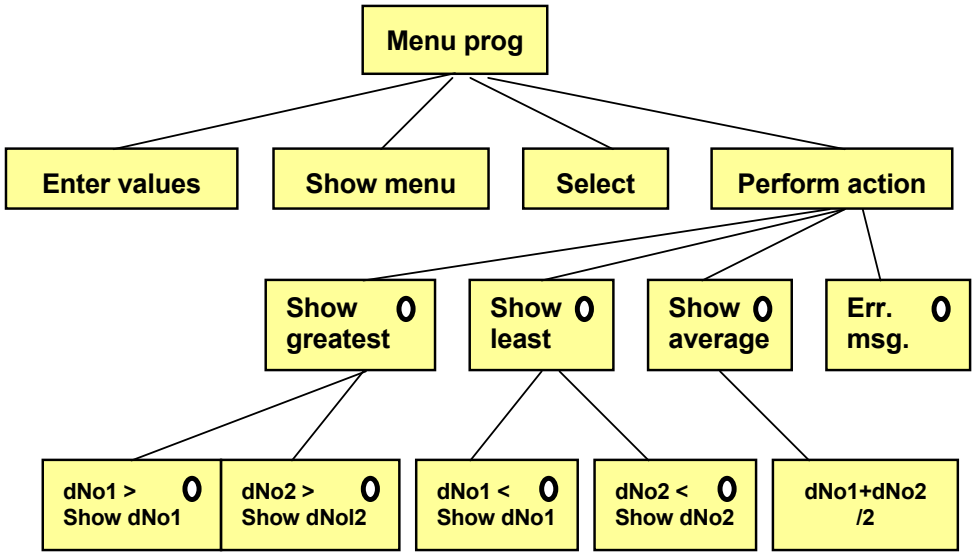
Since we have a selection situation where only one of the options should be performed, we indicate this with a circle in the upper right corner in each selection box.

The four options contain some logic, so we break down the JSP graph further:



In the "Show greatest" case we perform a check: if iNo1 is the greatest, we print it, otherwise we print iNo2. The "Show least" is analoguous. In the "Show average" case we add the two numbers and divide by 2.

The code will be this:

```
#include <stdlib.h>
#include <iostream.h>
void main()
{
  int iOpt;
  double dNo1, dNo2;
  cout << "Enter 2 numbers: ";
  cin >> dNo1 >> dNo2;
  system("cls");
  cout << "1.  Greatest" << endl;
  cout << "2.  Least" << endl;
  cout << "3.  Average" << endl;
  cout << endl << "Select: ";
  cin >> iOpt;
  switch (iOpt)
  {
    case 1:
      if (dNo1>dNo2)
```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